WELL-ORGANIZED COMMIT

This commit focuses on one change: implementing the sigmoid function and cross-entropy loss. This makes it easy to identify what was added and how the code was improved.

Update with sigmoid and cross entropy loss

main

1 file changed

commit message explains clearly the change that was made

```
31  -       # i is the index of a training example
    33  +   # Start building the total loss expression
    34  +   cost = 0
    35  +
    36  +   # Loop through each training example
32  37      for i in range(n):
    38  +       # Build z = b + w0*x0 + w1*x1 + ...
33  39          z = b
34  40          for j in range(k):
35  41              # weights[j] is the symbolic weight for feature j
36  42              # X[i, j] is the numerical value of feature j for training example i
37  -               z = z + weights[j] * X[i, j]
38  -               p = 1 / (1 + Variable.exp(-z))
    43  +           z = z + weights[j] * X[i, j]
    44  +
                # Apply sigmoid to get the predicted probability
                # logistic regression needs to convert z into a probability between 0 and 1
                p = 1 / (1 + Variable.exp(-z))

                # Calculate cross-entropy loss for this example
    50  +       loss = -(y[i] * Variable.log(p)
    51  +               + (1 - y[i]) * Variable.log(1 - p))
    52  +
    53  +       # Add this example's loss to the total
    54  +       cost = cost + loss
    55  +
    56  +   # Divide by the number of examples to get average loss
    57  +   cost = cost / n
```

added the sigmoid code here

added cross-entropy loss calculation here

Here, I made a commit after implementing a specific functionality. Rather than committing at an arbitrary point in my work, I committed after completing a meaningful update. I also made sure to keep the changes focused and logically related.

POORLY ORGANIZED COMMIT

This commit combines several unrelated changes. Rather than representing one specific, meaningful change, it bundles multiple functionalities into a single commit, making it harder to track the development of my project.

Update with predict function

main

3 files changed

commit message doesn't encompass all changes, also multiple files changed because unrelated updates were made into one commit

```
65  +       for iteration in range(self.iterations):
66  +           # Calling cost.grad(values) recursively calculates the partial de
67  +           # of the cost with respect to every weight and bias
68  +           grads = cost.grad(values)
70  +           # Update the bias using its gradient
71  +           values["b"] = values["b"] - self.learning_rate * grads["b"]
```

code for gradient descent training loop

```
82  +   # use the learned weights and bias to make predictions
83  +   def predict(self, X):
84  +       z = X @ self.w + self.b # @ is the numpy matrix multiplier
85  +
86  +       # apply Sigmoid function to convert the weighted sums into probabi
87  +       # use np.exp() instead of Variable.exp() because z is now a NumP
88  +       p = 1 / (1 + np.exp(-z))
89  +
90  +       # Convert probabilities into class labels
91  +       # p >= 0.5 will return Boolean values, .astype(int) converts the Boolean values into integers,
92  +       # 1 for True, 0 for False
93  +       # Essentially, return 1 for probabilities >= 0.5, otherwise return 0
94  +       return (p >= 0.5).astype(int)
```

adding to the predict function

variable_class.py

```
...   @@ -1,3 +1,9 @@
1   + # Name: Ava Cai
2   + # Date: 10/21/26
3   + # Assignment: Project 2
4   + # Purpose of the program: Variable Class
5   + # AI DECLARATION: AI was not used in the creation of this program.
```

updating file headers in a completely different file

This commit reflects making a commit at the end of a working session rather than after completing a specific functionality. Instead, I should have made separate commits for each added functionality (e.g. Add gradient descent training loop to fit, Add file headers, etc.) to keep my commit history more organized and easier to follow.